

Mathematics

A Cultural Heritage

Lecture 1

Introduction: Mathematics and Truth

Lecture Outline

In some previous lectures we have studied these aspects of **Numbers**:

- Historical development of number systems

- Various kinds of numbers

- Numbers at work

This lecture will talk about

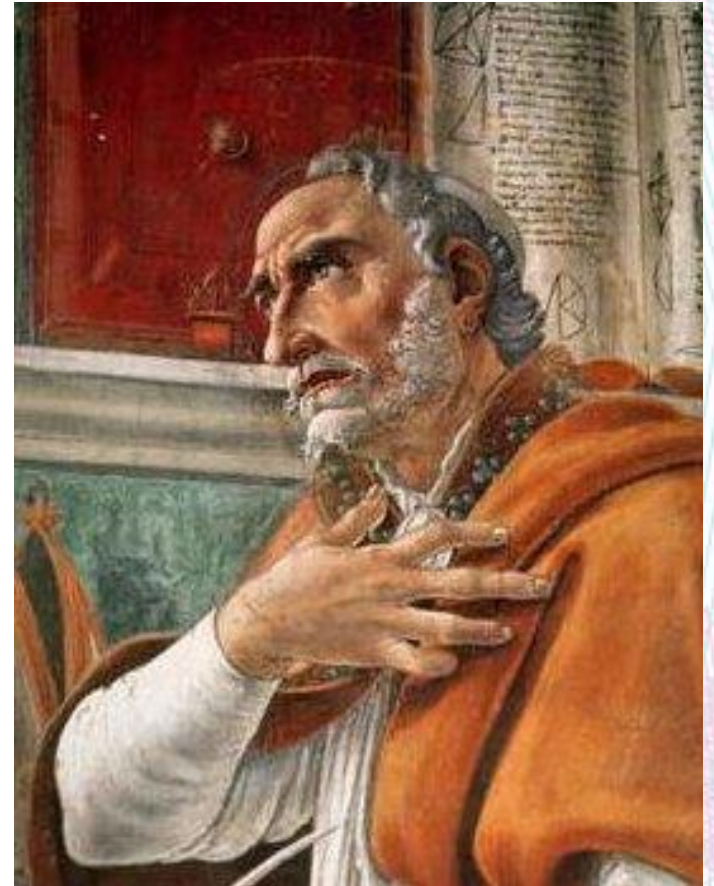
- Some of our **non-mathematical** perceptions of numbers, and

- Modern society's **obsessions** with numbers.

Perfect Numbers

“Six is a number perfect in itself, and not because God created all things in six days; rather, the converse is true. God created all things in six days because the number is perfect...”

– Saint Augustine, *The City of God*.



Saint Augustine (354-430)
Source: wikipedia

The Number 2

2 represents duality or dichotomies

Good and Evil

Yin and Yang

2 also represents repeating, or completeness

雙喜 (double happiness)

In middle ages Europe, Christianity viewed 2 as sinful.

In Cantonese dialects, 2 has connotation with “easy”.

28 is “easy money”

18.1 million HKD plate in 2016)

Missing Floors in Some Buildings



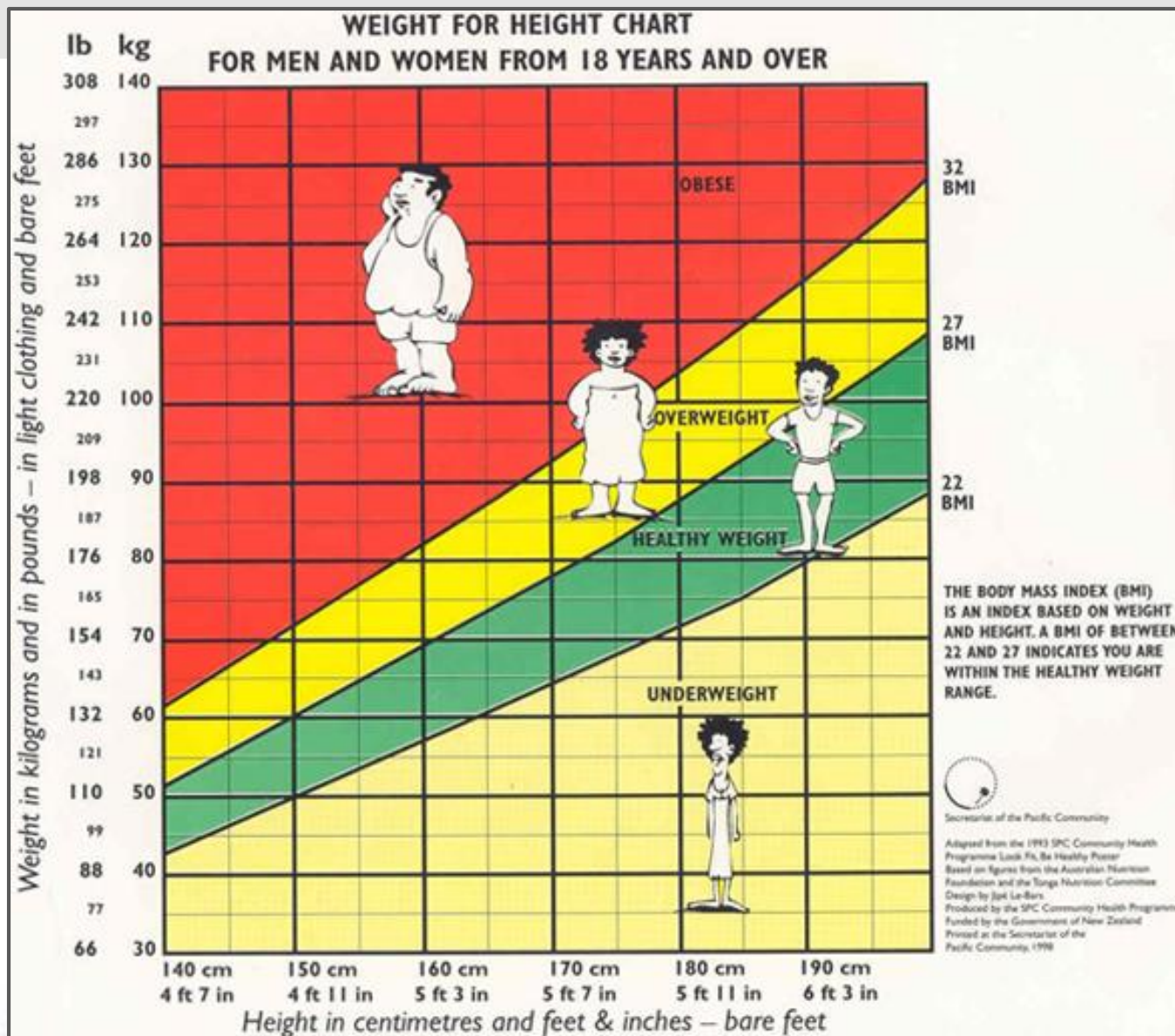
Gross National Product & Gross Domestic Product

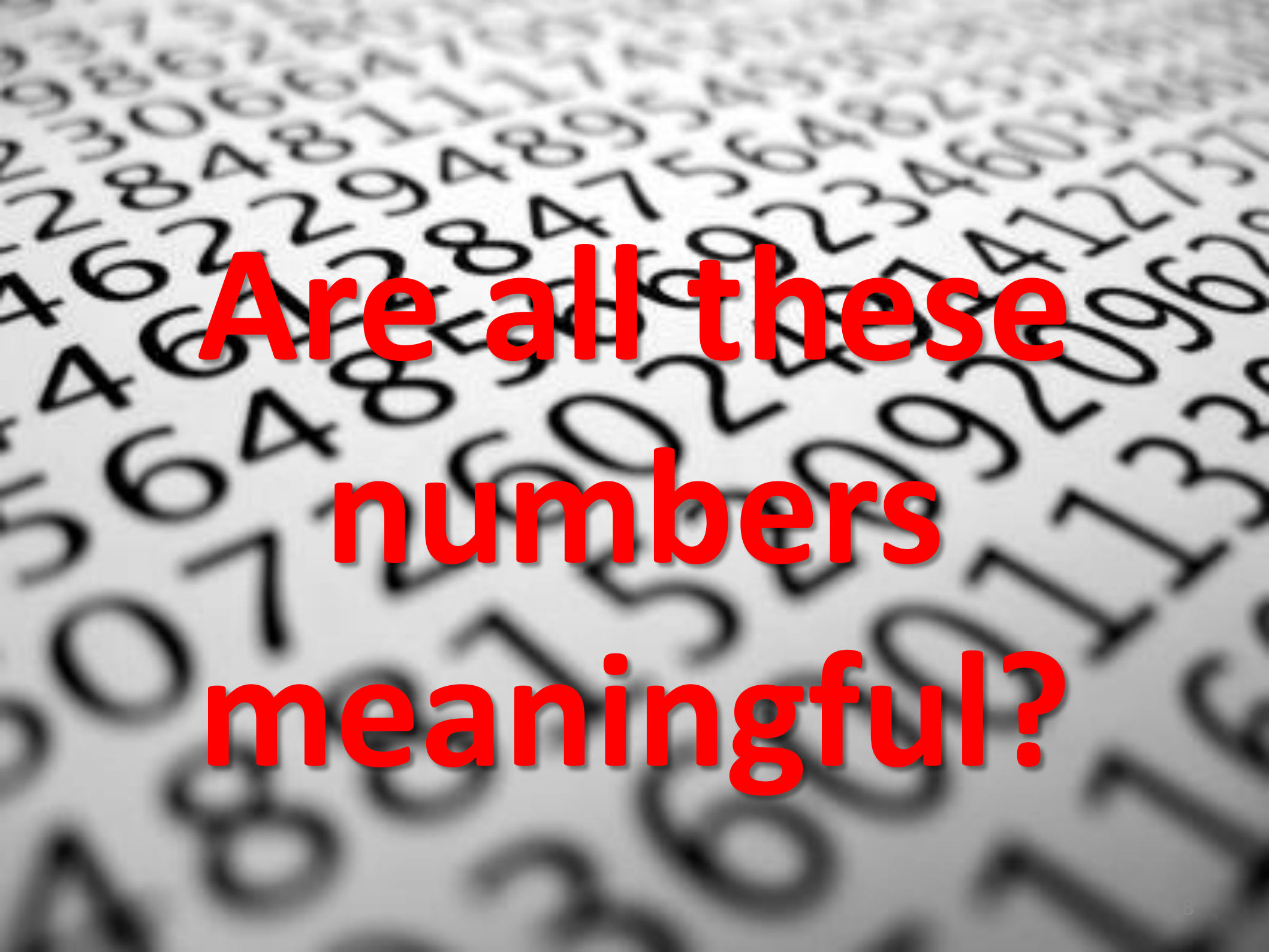
Rank ▲	2011 ◆		2010 ◆		2009 ◆	
1	 United States	15,097,083	 United States	14,648,955	 United States	14,135,520
2	 China	6,628,086	 China	5,717,592	 China	4,822,913
3	 Japan	5,774,376	 Japan	5,359,236	 Japan	4,793,538
4	 Germany	3,594,303	 Germany	3,513,807	 Germany	3,472,823
5	 France	2,775,664	 France	2,745,670	 France	2,742,735
6	 United Kingdom	2,366,544	 United Kingdom	2,373,636	 United Kingdom	2,532,124
7	 Italy	2,246,998	 Italy	2,149,222	 Italy	2,141,109
8	 Brazil	2,107,628	 Brazil	1,859,414	 Brazil	1,575,897
9	 India	1,746,481	 India	1,539,419	 Spain	1,469,901
10	 Canada	1,570,886	 Canada	1,475,865	 Canada	1,412,899

GNP of countries

Source: wikipedia

BMI



The background of the slide is a dense, out-of-focus field of black numbers (0-9) scattered across a light gray background. The numbers are of various sizes and orientations, creating a sense of randomness and data overload.

**Are all these
numbers
meaningful?**